

UPAY SHUROKKHA

Safer payments. Smarter agent liquidity.

AI Hackathon 2026 · DIU CPC × upay

Independent prototype · Synthetic data only

Decisions the prototype supports

For agents

Will I have enough cash and float for the next customer?

Forecast both demand streams.
Choose an affordable exchange.
Measure served demand and net costs.

For customers

What should I verify before sending?

Read the payment context.
Explain the warning signs.
Offer a safe next action and review.

Cash and float move in opposite directions

Customer cash-in: 1,000 BDT

Agent cash +1,000 | Agent float –1,000

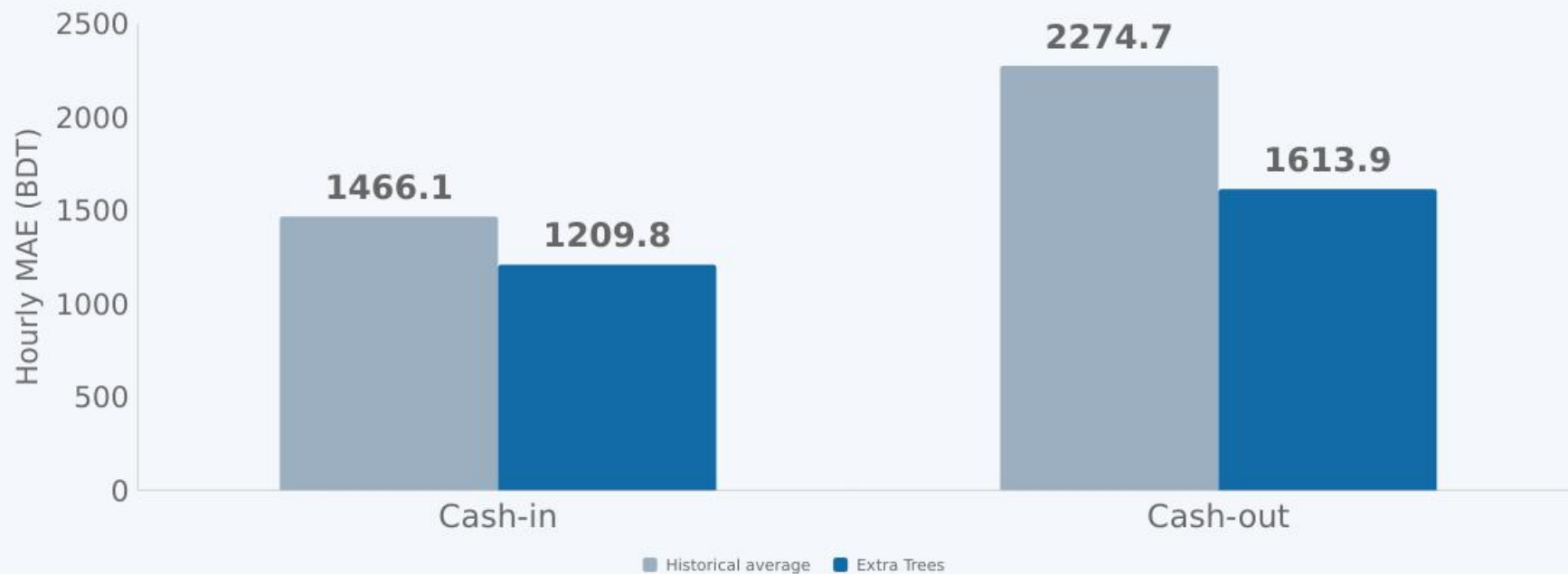
Customer cash-out: 1,000 BDT

Agent cash –1,000 | Agent float +1,000

An exchange changes the balance mix. Total working capital stays constant.

The optimizer includes no action and skips exchanges that become unaffordable.

Forecast error on unseen days



Lower is better · 3 agents × 21 test days × 12 hours · All demand is synthetic

Net earnings after exchange costs



+10.99 BDT vs historical forecasting; +5.22 BDT vs no action

Illustrative rates and capital · These outcomes do not establish commercial returns

Agent Planning: a clear next action

Inputs visible to agent

Counted physical cash + sample electronic float

Suggestion

Exchange an affordable amount before the busy hour, or keep balances.

Why: fictional transaction history, forecast cash-in/out, assumed exchange cost.

The actual synthetic day can be revealed for a same-demand comparison.

Payment Safety: explain each signal

Customer enters

Amount and optional reason for sending

Fictional history provides

Usual amount and whether this is a first payment

Output: concern, reasons, independent verification and a demo choice.

Text classifier: useful signal, small test



84.6% precision · 91.7% recall · 16.7% false-warning rate

Only 24 test sentences, eight per language · Full concern score is not calibrated

Limits that shape the next experiment

Liquidity

Refused demand may be missing from real histories.
Partner availability, travel time and financing need modeling.

Payment safety

Negation and paraphrases can cause mistakes.
New-account and language cohorts need much larger tests.

Pilot first in shadow mode, then test human-approved actions.

Measure full net costs, missed demand, false warnings, reviewer effort and comprehension.

Working prototype and Phase 2 path

Built and reproducible

Application, synthetic data, models, policy, evaluation,
15 automated tests, report and demonstration script.

Adaptable for on-site requirements

Change capital and costs, add partner constraints,
update forecasts or revise the review policy.